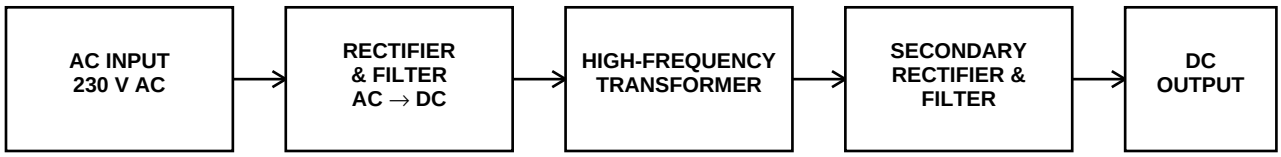
# SMPS and UPS – Power Supply Assignment

## 1. Real-World Devices Using SMPS and UPS

| Type | Device | Why it needs this power supply |
|---|---|---|
| **SMPS** | Desktop computer | Converts AC mains into stable DC voltages required by the motherboard, CPU, drives and oth |
| **SMPS** | LED/LCD monitor | Converts mains electricity into the low DC voltages needed by the display electronics. |
| **SMPS** | Wi-Fi router | Provides regulated low-voltage DC power efficiently from household AC power. |
| **UPS** | Desktop PC | Keeps the computer powered temporarily during a power cut and helps prevent sudden shutd |
| **UPS** | Network router/modem | Provides backup power during outages so internet/network equipment can continue operating |

## 2. Labeled Diagram: Typical Desktop PC SMPS




Typical flow: AC mains → rectification/filtering → high-frequency switching/transformer → secondary rectification/filtering → regulated DC outputs (e.g., +12 V, +5 V, +3.3 V).

## 3. Common PC Power Connectors

| Connector | Devices/components powered | Safety tip |
|---|---|---|
| **ATX 24-pin motherboard connector** | Main motherboard power. | Switch off and unplug the PC before connecting/disconnecting; align t |
| **SATA power connector** | SATA HDDs, SSDs and some optical drives. | Insert it in the correct orientation and disconnect power before removi |
| **Molex connector** | Older IDE/PATA drives, optical drives, fans and some peripherals. | Hold the plastic connector rather than pulling the wires, and power off |
| **PCIe 6/8-pin connector** | Dedicated graphics cards (GPUs). | Use the PSU's PCIe cable, not a CPU/EPS cable; ensure the clip lock |

## Connector Identification – simple labeled reference diagrams

| | | | |
|---|---|---|---|
| **ATX 24-pin**<br>■ ■ ■ ■ ■ ■<br>■ ■ ■ ■ ■ ■<br>■ ■ ■ ■ ■ ■ ■ ■ | **SATA Power**<br>■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■<br>Flat, wide connector with L-shaped key | **Molex**<br>● ● ● ●<br>4-pin peripheral power connector | **PCIe 6/8-pin**<br>■ ■ ■<br>■ ■ ■ ■<br>GPU power connector |

Note: The connector sketches above are schematic identification diagrams, not photographs. Actual connector shapes may vary slightly by manufacturer.

## 4. Connecting an SMPS in a New Gaming PC

**1. Prepare the PSU:** Mount the SMPS/PSU securely in the PC case and make sure the mains switch is OFF and the AC cable is unplugged.

**2. Motherboard:** Connect the large **24-pin ATX motherboard cable** to the motherboard's 24-pin ATX socket. Make sure the locking clip clicks into place.

**3. CPU power:** Connect the PSU's **4+4-pin EPS/CPU cable** to the motherboard CPU power socket near the processor. This is separate from the PCIe GPU cable.

**4. Graphics card:** Connect the required **PCIe 6-pin, 8-pin, or 6+2-pin cable** from the PSU to the graphics card. Use the number of connectors specified by the GPU.

**5. Storage:** Connect a **SATA power connector** from the PSU to each SATA HDD/SSD or SATA optical drive. (An M.2 NVMe SSD normally does not need a separate SATA power cable.)

**6. Accessories:** If an older fan or accessory requires a **Molex connector**, connect the appropriate Molex cable without forcing it.

**7. Final check:** Check that every connector is fully seated, cables are clear of fans, and no cable is being pulled or sharply bent.

**8. Power on:** Connect the AC power cable, turn the PSU switch ON, and then start the PC. If the system does not start normally, switch it off and re-check the connections.

**Conclusion:** An SMPS efficiently converts AC power into regulated DC power for computer components, while a UPS provides temporary backup power during interruptions. Correct connector selection and safe handling are essential when building or servicing a PC.